Team Number: C- _____ Team Name: _____
Individual Names: _____
Final Score: ____ /100

Thermodynamics(C)

Clements Invitational, December 16th 2017

- Keep this test closed until told to start by the proctor.
- Bubble multiple choice answers in Scantron
- Write team number/name on every page. / Staple all # pages together.
- Use correct significant figures when answering. No credit will be given to answers with incorrect significant figures.
- Each question is worth 1 point unless otherwise noted.
- Take a deep breath (required for credit)

1 Thermodynamics

Use the following information for the first 7 questions.

I. Isothermal

II. Adiabatic

III. Isobaric

IV. Isochoric

1.1. Which process(es) occur/occurs at a fixed pressure?

A) **I.**

B) **II.**

C) **III.**

D) **III.** and **IV.**

1.2. Which process(es) occur/occurs at a fixed volume?

A) **I.**

B) **II.**

C) **IV.**

D) **II.** and **IV.**

1.3. In which process(es) is the work done equal to 0?

A) **I.**

B) **II.**

C) **III.**

D) **IV.**

1.4. In which process(es) is the net heat exchanged with the environment equal to 0?

A) **I.**

B) **II.**

C) **III.**

D) **I.** and **II.**

1.5. In which process(es) is the entropy change equal to 0?

A) **I.**

B) **II.**

C) **IV.**

D) **II.** and **III.**

1.6. In which process(es) is the work done equal to the product of pressure and the change in volume?

A) **I.**

B) **II.**

C) **III.** and **IV.**

D) Work=$P \times \Delta V$ in all processes

1.7. In which type(s) of process(es) occur/occurs during the Carnot cycle?

A) **I.**

B) **II.**

C) **I.** and **II.**

D) **I.** and **IV.**

1.8. In an isothermal process, which of the following relations must be true?

A) $P_1V_1 = P_2V_2$

B) $q = P_2V_2$

C) $w = P_2V_2$

D) $w = 0$

1.9. A gas undergoes an isothermal process. By what amount does the internal energy of the gas change?

A) $P \times \Delta V$

B) 0

C) Cannot be determined without knowledge of the temperature

D) Cannot be determined without knowledge of how many moles of gas there are

1.10. Complete the statement: The Principle of Equipartition of Energy states that energy is shared equally amongst

A) The quantum states of the system

B) The degrees of freedom

C) Each molecule

D) Each different form of energy except kinetic energy

1.11. What property allows the calculation of temperature change of a substance with respect to the amount of heat added?

A) Density

B) The Partition function

C) Entropy

D) Specific heat

1.12. Which of the following is the SI unit of temperature measurement?

A) Kelvin

B) Celsius

C) Farenheit

D) Rankine

1.13. Which of the following is the SI unit of heat associated with melting?

A) Latent Heat of vaporization

B) Free energy of fusion

C) Latent heat of fission

D) Latent heat of fusion

1.14. At which point on a phase diagram can all the phases of a substance be observed?

A) Critical point

B) Melting point

C) Triple point

D) Omnipoint

Convert each of the following temperatures from Farenheit to Celsius.

1.15. -40 F=_____C

1.16. 0 F=_____C

1.17. 40 F=_____C

1.18. 32 F=_____C

1.19. 212 F=_____C

1.20. A student wants to determine the heat of dissolution for common table salt. He dissolves 23 grams of table salt in a coffee cup of water(100 mL) and measures the temperature of the water. He finds the temperature drops by about 7.5 degrees Celsius. Calculate the heat of dissolution for NaCl Assume table salt is 100% NaCl. Show all work associated.(4 points)

Match each of the following scientists to the thermodynamic concept they are associated with:

A) Created the original formulation of statistical thermodynamics and the original mathematical definition of entropy

B) Provided an upper limit on the the efficiency of a classical thermodynamic system

C) Gave the first ideas behind the second law of thermodynamics, and first introduced the idea of entropy

D) Gave the first notion that heat could not be converted back to mechanical energy, which gave rise to the second law. Also credited with creating the absolute temperature scale used by the SI system

E) One of the first to apply thermodynamics to chemistry, and gave the relation between the first law and entropy

1.21. _ Josiah Gibbs

1.22. _ Rudolf Clausius

1.23. _ Ludwig Boltzmann

1.24. _ William Thompson

1.25. _ Sadi Carnot

2 A really cool engine!!

A Mad scientist has designed a new thermodynamic engine. He wants to calculate the work and heat lost and gained in his cycle, so he can compute the efficiency of his cycle. Unfortunately, he has lost his table of state variables which he has measured! He knows the following details about his cycle (its long)

His engine consists of 1 mole of a monatomic gas taken through the following scenarios. The initial conditions are 100 atmospheres, .5 liters.

- Isobaric expansion to a volume of 1 L
- Isochoric cooling to a temperature of 853.06 K
- Isobaric expansion to a volume of 1.5 L
- Isochoric heating to a pressure of 100 atm
- Isobaric expansion to a volume of 2 L
- Isochoric cooling to a temperature of 243.73 K
- Isobaric compression to a volume of 1 L
- Adiabatic heating to a pressure of 75 atm
- Isothermal expansion to a volume of .5 L
- Isochoric expansion back to the initial conditions

(5 points) Sketch a PV diagram of the above described engine. While you need not worry about scaling, you should worry about relative placement of values (for example, a pressure of 100 atmospheres must be located higher than a pressure of 75 atmospheres, but you need not worry about how much higher)

In the following table, let 1 denote the initial state of the system, and 10 denote the last state of the system before it returns to initial conditions. Fill in all missing variables. Make sure to use the correct signs. 1 point per blank.

Variables	1	2	3	4	5	6	7	8	9	10
P(atm)	100.00	100.00			100.00	100.00			75.00	
V(L)	.500	1.00		1.500		2.00		1.00		.500
T(K)			853.06				243.73			

Calculate the amount of work done by the system in each process, as well as the amount of heat added or released in each process. 2 points per blank

Step	Work(J)	Heat(J)
1		
2		
3		
4		
5		
6		
7		
8		
9		
10		

Calculate the efficiency of the engine. 10 points.

Clements Invitational - Repaired practice edition

Division C | Thermodynamics | Repair appendix

This appendix contains authored replacements and supplied written alternatives. It does not reconstruct unpublished historical material. For the listed item IDs, use this appendix instead of the archived wording and key.

Name: _____ Date: _____

1-13. (1 points) Authored replacement for source item Question 1-13: What is the SI unit of heat energy needed to melt a sample? Distinguish it from the unit of specific latent heat of fusion.

Answer: _____
_____
_____

1-20. (6 points) Authored replacement for source item Question 1-20: Solve this case and explain your reasoning: A system receives 500 J heat and does 200 J work. Find delta U and chemistry work.

Answer: _____
_____
_____